Heterogeneity in Dynare 7
Describing the model

Dynare Masterclass 2026

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Roadmap

1. **Problem**: the abstract framework and Krusell–Smith.
2. **Syntax**: the `.mod` file.
3. **Restrictions and specifications**: parser rules.
4. **Under the hood**: what the preprocessor generates.

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The abstract framework: heterogeneous block

As in Bhandari et al. [2023], a continuum of agents $i \in \mathcal{I}$, discrete time.
Red: heterogeneous. Blue: aggregate.
Heterogeneous block:

$$F(\theta_{i,t}, a_{i,t-1}, x_{i,t}, \mathbb{E}_{i,t} x_{i,t+1}, Y_t) = 0.$$

where

- $x_{i,t}$: heterogeneous endogenous variables
- $a_{i,t-1} \subseteq x_{i,t-1}$: heterogeneous pre-determined variables
- $\theta_{i,t}$: heterogeneous exogenous variables
- Y_t: relevant aggregate variables
- $\dim F = \dim x_{i,t}$:

The abstract framework: aggregate block

Aggregate block $\approx$ standard RA block

Aggregate block:

$$G\left(\int y_{i,t}\, di,\ A_{t-1},\ X_t,\ \mathbb{E}_t X_{t+1},\ \mathscr{E}_t\right) = 0.$$

- X_t: aggregate endogenous variables
- $A_{t-1} \subseteq X_{t-1}$: pre-determined variables.
- $\mathscr{E}_t$: aggregate i.i.d Gaussian innovations
- $\dim G = \dim X_t$

Relationship with the heterogeneous block:

- $y_{i,t} \subseteq x_{i,t}$: relevant aggregated heterogeneous variables
- $Y_t \subseteq [A_{t-1}, X_t, \mathbb{E}_t X_{t+1}, \mathscr{E}_t]$.
- $\theta_{i,t}$ Markovian, independent across i, independent of $\{\mathscr{E}_t\}$.

Running example: Krusell and Smith (1998)

General description

- Continuum of infinitely-lived households $i \in [0, 1]$.
- Inelastic labor supply.
- Self-insurance against idiosyncratic productivity risk, incomplete markets.
- Representative firm with Cobb–Douglas technology.
- Aggregate uncertainty: AR(1) TFP shock Z_t.
- Idiosyncratic productivity independent of Z_t (unlike Krusell and Smith [1998]).

Running example: Krusell and Smith (1998)

Household problem and FOCs

Program:

$$V(k_{i,t-1}, e_{i,t}) = \max_{c_{i,t}, k_{i,t}} \left\{ U\left(c_{i,t}\right) + \beta \mathbb{E}_{i,t} V(k_{i,t}, e_{i,t+1}) \right\}.$$

$$\text{s.t.} \begin{cases} U\left(c_{i,t}\right) = \dfrac{\left(c_{i,t}\right)^{1-1/\eta}}{1-1/\eta}, k_{i,t} \geq 0 \\ e_{i,t} = \rho^e e_{i,t-1} + \epsilon^e_{i,t} \\ c_{i,t} + k_{i,t} = (1 + r_t) k_{i,t-1} + w_t \exp\left(e_{i,t}\right) \end{cases}$$

First Order Conditions → F

$$k_{i,t} \geq 0 \quad \perp \quad \left(c_{i,t}\right)^{-1/\eta} - \beta \mathbb{E}_{i,t}\left[(1 + r_{t+1}) \left(c_{i,t+1}\right)^{-1/\eta}\right] \geq 0$$

$$c_{i,t} + k_{i,t} = (1 + r_t) k_{i,t-1} + w_t \exp\left(e_{i,t}\right)$$

Running example: Krusell and Smith (1998)

Aggregate block

Aggregate optimality conditions $\to G$

$$
\begin{aligned}
Y_t^a &= \exp(Z_t)\, K_{t-1}{}^{\alpha} L^{1-\alpha}, \\
r_t &= \alpha \exp(Z_t)\, K_{t-1}{}^{\alpha-1} - \delta, \\
w_t &= (1-\alpha) \exp(Z_t)\, K_{t-1}{}^{\alpha}, \\
K_t &= \int k_{i,t}\, di, \\
\log Z_t &= \rho^Z \log Z_{t-1} + (1-\rho^Z) \log Z_{ss} + \epsilon_t^Z.
\end{aligned}
$$

Mapping Krusell–Smith to the abstract framework

Heterogeneous side:

$$x_{i,t} = [c_{i,t},\ k_{i,t}]', \quad a_{i,t-1} = [k_{i,t-1}]',$$
$$\theta_{i,t} = [e_{i,t}]', \quad Y_t = [r_t,\ w_t]'.$$

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Aggregate side:

$$X_t = [Z_t,\ K_t,\ r_t,\ w_t,\ Y_t^a]',$$
$$A_{t-1} = [K_{t-1},\ Z_{t-1}]', \quad \mathcal{E}_t = [\epsilon_t^Z]'.$$

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Only $k_{i,t}$ enters G via an integral: $y_{i,t} = [k_{i,t}]'$.

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  $\rightarrow$ `heterogeneity_dimension`.
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  → `var(heterogeneity=...)`, `var`.
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- Individual exogenous $\theta_{i,t}$
  → `varexo(heterogeneity=...)`.
  KS: $\theta_{i,t} = [e_{i,t}]'$.
- Aggregate innovations $\mathscr{E}_t$
  → `varexo` and the `shocks` block.
  KS: $\mathscr{E}_t = [\epsilon^Z_t]'$.
- Parameters: `parameters`. No heterogeneity option yet.

Dynare .mod file

Heterogeneous variables

```
heterogeneity_dimension households;

var(heterogeneity=households)
    c  (long_name = 'Consumption')
    k  (long_name = 'Assets')
;

varexo(heterogeneity=households)
    e  (long_name = 'Idiosyncratic productivity shock')
;
```

Dynare .mod file

Aggregate variables

```
var
    Y (long_name = 'Aggregate output')
    r (long_name = 'Rate of return on capital net of depreciation')
    w (long_name = 'Wage rate')
    K (long_name = 'Aggregate capital')
    Z (long_name = 'Aggregate productivity shock')
;

varexo eps_Z (long_name = 'Aggregate productivity innovation');
```

Dynare .mod file

Parameters and aggregate innovations

```
parameters
 L     (long_name = 'Labor')
 alpha (long_name = 'Share of capital in production function')
 beta  (long_name = 'Subjective discount rate of households')
 delta (long_name = 'Capital depreciation rate')
 eis   (long_name = 'Elasticity of intertemporal substitution')
 rho_Z (long_name = 'Aggregate TFP shock persistence')
 sig_Z (long_name = 'Aggregate TFP shock innovation std err')
 Z_ss  (long_name = 'Aggregate TFP shock average value')
;

L = 1; Z_ss = 0.8816460975214567; alpha = 0.11;
beta = 0.9819527880123727; delta = 0.025; eis = 1;

shocks;
    var eps_Z; stderr 0.01;
end;
```

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Equations

Individual conditions:

$$F(\theta_{i,t}, a_{i,t-1}, x_{i,t}, \mathbb{E}_{i,t} x_{i,t+1}, Y_t) = 0.$$

→ `model(heterogeneity=...)` block.

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$$G\left(\int y_{i,t}\, di,\ A_{t-1},\ X_t,\ \mathbb{E}_t X_{t+1},\ \mathscr{E}_t\right) = 0.$$

→ `model` block.

→ `SUM(x)` $\approx \int x_{i,t} di$ operator to sum x on the distribution of agents. Inside aggregate equations only!

Dynare .mod file

Heterogeneous equations block

```
model(heterogeneity=households);
    [name='Euler with borrowing constraint']
    c^(-1/eis) - beta*(1+r(+1))*c(+1)^(-1/eis) = 0 ⟂ k >= 0;

    [name='Budget constraint']
    c + k = (1+r)*k(-1) + w*exp(e);
end;
```

Dynare .mod file

Aggregate equations block

```
model;
    [name='Production function']
    Y = exp(Z) * K(-1)^alpha * L^(1 - alpha);
    [name='Capital rental rate']
    r = alpha * exp(Z) * (K(-1) / L)^(alpha - 1) - delta;
    [name='Wage rate']
    w = (1 - alpha) * exp(Z) * (K(-1) / L)^alpha;
    [name='Capital market clearing']
    K = SUM(k);
    [name='TFP law of motion']
    log(Z) = rho_Z*log(Z(-1)) + (1-rho_Z)*log(Z_ss) + eps_Z;
end;
```

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With RESID = LHS − RHS and x the bounded variable:

$$\begin{aligned} x = LB \quad &\Rightarrow \quad RESID(x) \geq 0, \\ LB < x < UB \quad &\Rightarrow \quad RESID(x) = 0, \\ x = UB \quad &\Rightarrow \quad RESID(x) \leq 0. \end{aligned}$$

Wrong sign?

- steady state → fails: the regime is picked from the inequality branch.
- dynamics → works: the interior branch $RESID = 0$ is sign-blind.

Implicit conditional expectations

- Standard Dynare: expectation *outside*.

$$\mathbb{E}_{i,t} F(\theta_{i,t}, a_{i,t-1}, x_{i,t}, x_{i,t+1}, \ldots) = 0.$$

- Here: expectation *inside*, on heterogeneous forward-looking variables.

$$F(\theta_{i,t}, a_{i,t-1}, x_{i,t}, \mathbb{E}_{i,t} x_{i,t+1}, Y_t) = 0.$$

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$$F(\theta_{i,t}, a_{i,t-1}, x_{i,t}, \mathbb{E}_{i,t} x_{i,t+1}, Y_t) = 0.$$

→ The pre-processor considers two regimes for leads:

- $\mathbb{E}_{i,t} x_{i,t+1}$ **stochastic:** a genuine expectation, integrating the idiosyncratic shock kernel. Idiosyncratic risk is priced.
- X_{t+1} **perfect foresight**: the dynamics solve for the linearized response after a one-time MIT shock. Plain lead, no expectation operator.
- Non-linear $\phi(x_{i,t+1})$: pre-processor introduces $z_{i,t} \equiv \phi(x_{i,t})$ and substitutes the affine lead $z_{i,t+1}$.

Non-separability of leads and lags

Heterogeneous variables: a lead $x_{i,t+1}$ and a lag $x_{i,t-1}$ entangled inside a non-separable operator are **rejected at parse time**.

- Rejected:

$$\log\left(x_{i,t-1} + x_{i,t+1}\right), \quad \exp\left(x_{i,t-1} \cdot x_{i,t+1}\right), \quad \left(x_{i,t-1} + x_{i,t+1}\right)^{\alpha}.$$

- Admissible (separable):

$$\beta\, x_{i,t+1} + (1 + r_t)\, x_{i,t-1}, \quad x_{i,t-1} \cdot c_{i,t+1}, \quad \left(x_{i,t-1}\right)^{\alpha} x_{i,t+1}.$$

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Separable: $+$, $-$ always; $\times$, $\div$ when one factor carries no lead.
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Separable: $+$, $-$ always; $\times$, $\div$ when one factor carries no lead.
→ The heterogeneous lead sits under $\mathbb{E}_{i,t}$: a non-separable coupling would need a nested, second-pass expectation. Not implemented yet.
Aggregate variables escape the rule: no expectation. Non-separable couplings allowed, as standard in Dynare.

Worked example: auxiliary variable in the Euler equation

Euler written with c alone:

$$\left(c_{i,t}\right)^{-1/\eta} - \beta \mathbb{E}_{i,t}\left[(1 + r_{t+1})\left(c_{i,t+1}\right)^{-1/\eta}\right] = 0 \quad \perp \quad k_{i,t} \geq 0.$$

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The heterogeneous lead $c_{i,t+1}$ enters non-linearly. The pre-processor introduces a contemporaneous auxiliary and replaces the non-linear lead by an affine one:

$$v_{i,t} \equiv \left(c_{i,t}\right)^{-1/\eta}, \qquad \left(c_{i,t}\right)^{-1/\eta} - \beta \mathbb{E}_{i,t}\left[(1 + r_{t+1})\, v_{i,t+1}\right] = 0.$$

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- The lead is now affine: $(1 + r_{t+1})\, v_{i,t+1}$, an aggregate factor times the heterogeneous lead $v_{i,t+1}$, separable.
- $v_{i,t}$ joins $x_{i,t}$; generated name AUX_HET_ENDO_LEAD.

Block squareness and admissible leads/lags

Squareness. Each block must be square:

$$\dim(F) = \dim(x_{i,t}), \quad \dim(G) = \dim(X_t).$$

Auxiliary variables count on both sides.
→ No additional equations allowed for now (e.g. Walras' law cross-checks)

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Leads/lags.

- Heterogeneous endogenous: lag $\in \{-1, 0\}$, lead $\in \{0, +1\}$.
- Heterogeneous exogenous: lag $= 0$ only.
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→ Two-period-ahead heterogeneous lead? Introduce $\tilde{x}_{i,t} \equiv x_{i,t+1}$ by hand.

Specification of the exogenous shocks

Idiosyncratic shocks $\{\theta_{i,t}\}_{i,t}$ must be **independent Markov processes**:

- i.i.d. across agents i;
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Ruled out:

- Cross-agent correlation
  $\rightarrow$ A shock common to all agents belongs in X_t, not $\theta_{i,t}$.
- Idiosyncratic-aggregate correlation

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Heterogeneous block F, prefix dynamic_het1_ (het1: the heterogeneity dimension):

- resid, g1, g2: residuals, Jacobian, Hessian of F.
- set_auxiliary_variables: preprocessor-introduced auxiliaries.
- complementarity_conditions: the (LB, UB) MCP pair.

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- `resid`, `g1`, `g2`: residuals, Jacobian, Hessian of F.
- `set_auxiliary_variables`: preprocessor-introduced auxiliaries.
- `complementarity_conditions`: the (LB, UB) MCP pair.

Aggregate block G, prefix `dynamic_`.

- `resid`, `g1`, `g2`: residuals, Jacobian, Hessian of G.
  → an extra `yagg` argument feeds the cross-sectional integrals $\int y_{i,t}\, di$.

Auxiliary variables for complementarity

Euler equation

Euler after the ν auxiliary:

$$c_{i,t}^{-1/\eta} - \beta \mathbb{E}_{i,t}\left[(1 + r_{t+1}) \nu_{i,t+1}\right] = 0 \quad \perp \quad k_{i,t} \geq 0.$$

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Preprocessor introduces slack $\mu_{i,t}$ (internal name `MULT_L_k`)
→ in `dynamic_het1_resid.m`:

$$\begin{aligned}
&\left(c_{i,t}{}^{-1/\eta} - \beta \mathbb{E}_{i,t}\left[(1 + r_{t+1})\, v_{i,t+1}\right]\right) - \mu_{i,t} = 0,\\
&k_{i,t} \cdot \mu_{i,t} = 0.
\end{aligned}$$

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→ assigned in `dynamic_het1_set_auxiliary_variables.m`:

$$\mu_{i,t} = \left(c_{i,t}{}^{-1/\eta} - \beta \mathbb{E}_{i,t}\left[(1 + r_{t+1})\, \nu_{i,t+1}\right]\right) \cdot \mathbf{1}_{\{k_{i,t} \leq 0\}}.$$

Aggregated heterogeneous variables

Needed for market clearing conditions, e.g.

$$K_t = \int k_{i,t} \, di$$

The corresponding syntax is

```
K = SUM(k)
```

→ The pre-processor

- creates auxiliary aggregate variables (e.g. `SUM_k`)
- substitutes those auxiliary aggregate variables via `yagg`

Bibliography

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